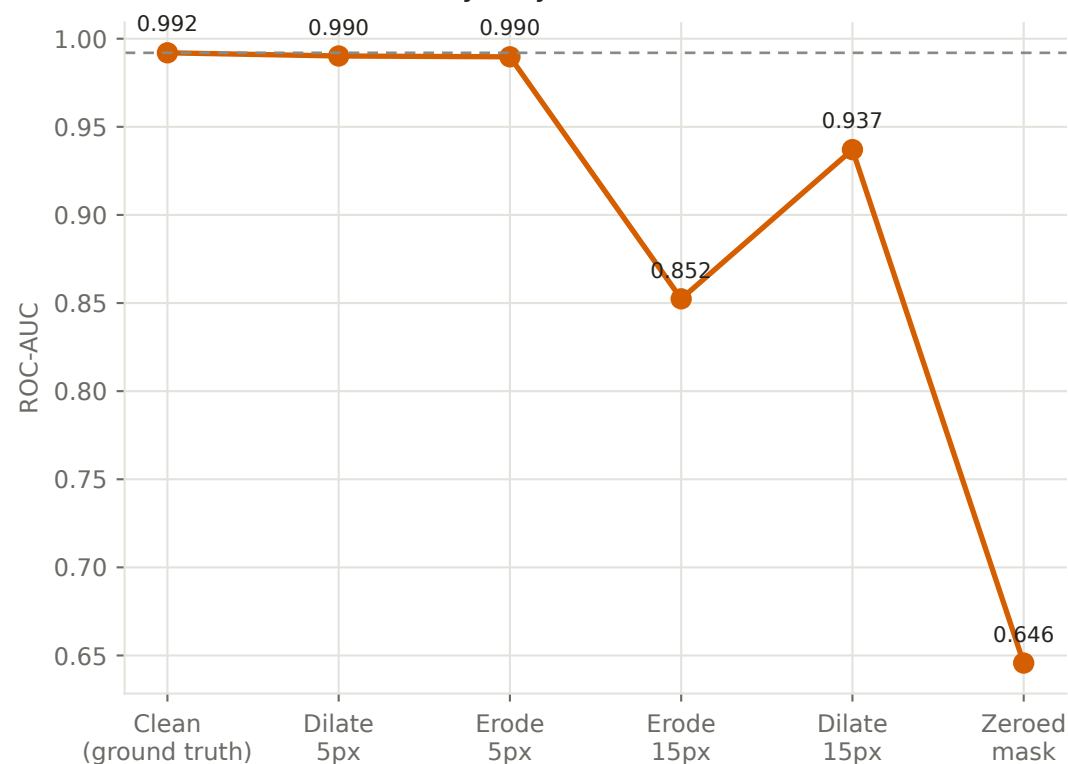
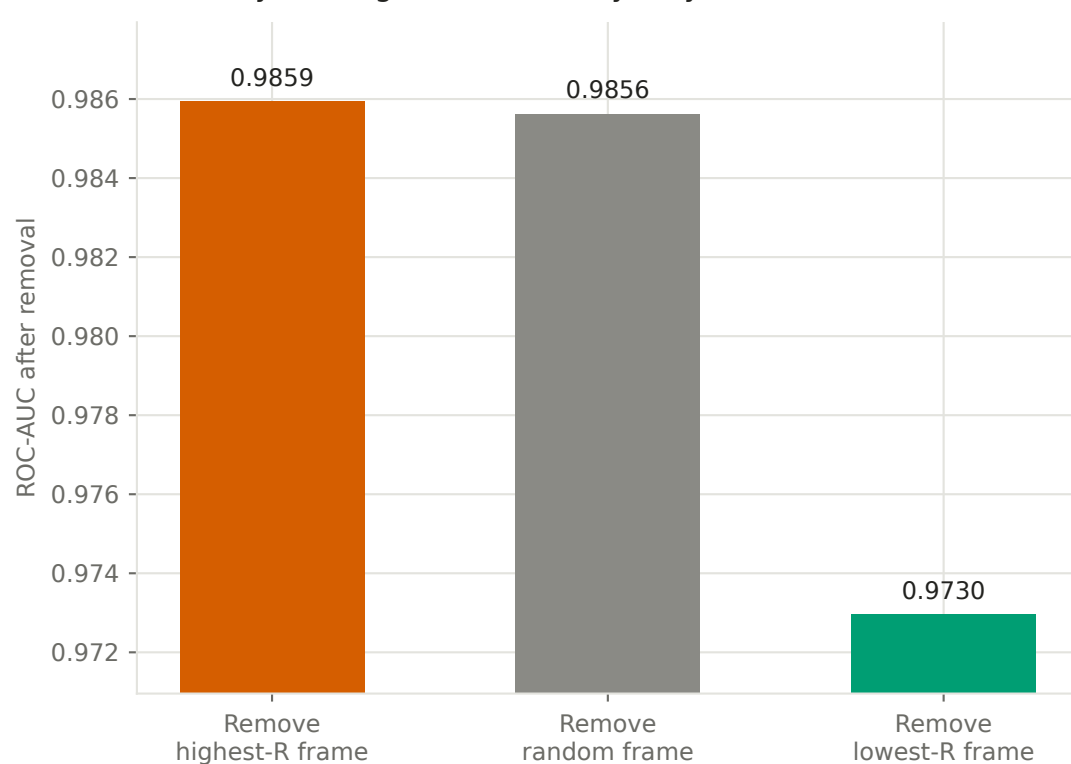


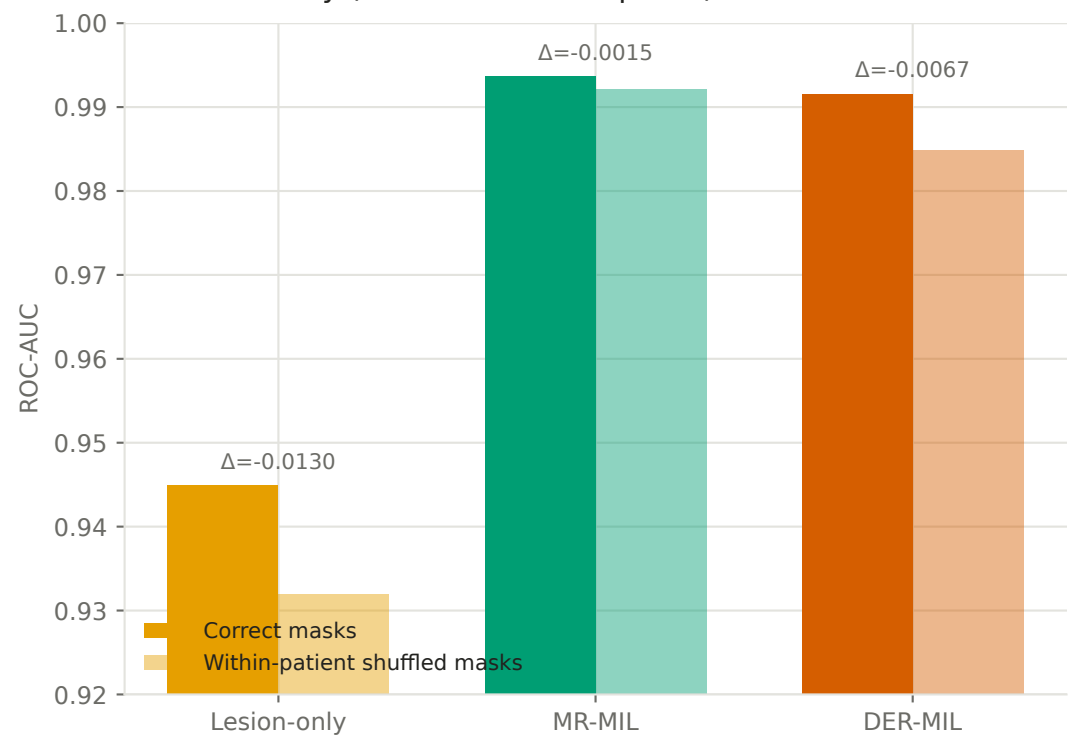
(a) Segmentation-quality degradation, no retraining
DER-MIL, ConvNeXt-Tiny, ThyroidXL test



(b) One frame removed, by DER-MIL's own reliability ranking (ConvNeXt-Tiny, ThyroidXL test)



(c) Within-patient mask-identity shuffle
ConvNeXt-Tiny (shortcut-reliance probe)



(d) Frame-order permutation invariance
(DER-MIL, ConvNeXt-Tiny)

Patients tested	707
Mean $ \Delta $ predicted probability	0.00e+00
95th-percentile $ \Delta p $	0.00e+00
Max $ \Delta p $ across all patients	0.00e+00
Shuffling the order in which a patient's frames are fed to the model changes its prediction by less than 1e-6 on this backbone too: the aggregation stage is exactly permutation invariant by construction.	